New Long-Range Interactions Affect Neutrinos Inside Earth

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Motivation

Neutrinos offer a rare glimpse into physics beyond the Standard Model. Long-range interactions, from symmetries like $L_e - L_\mu$, can measurably affect oscillations via Earth's electron content. As experiments grow more precise, exploring these effects is important to uncover testable new physics. This work models such interactions using realistic Earth profiles and numerical methods to study their implications in neutrino oscillation experiments.

Objectives

- Compute the effect of long-range neutrino interactions, sourced by all electrons in the Earth, on neutrino oscillations using realistic Earth models.
- Solve the Schrödinger equation numerically to evaluate how these interactions modify oscillation probabilities across different energies and trajectories.

Long-range Interaction

A new ultra-light boson Z', associated with the flavor symmetry $L_e - L_\mu$, mediates a long-range interaction between neutrinos and electrons. This interaction is characterized by a coupling g' and a boson mass $m_{Z'}$.
As an example, setting $g' = 10^{-26}$ and $m_{Z'} = 10^{-13} eV$ gives an interaction range:

$$r_{Z'} \sim \frac{1}{m_{Z'}} \approx 6371\,\mathrm{km}$$

which corresponds to the Earth's radius. These values are not fixed, but chosen to illustrate a scenario where the entire Earth contributes to the potential. It is associated with the interaction with all the electrons within the Earth, and its contribution at the neutrino position is:

$$V_{\mathrm{BSM}} = \frac{g'^2}{4\pi} \int_{V_{earth}} d^3r \, \sin(\theta) n_e(r) \frac{e^{-mr}}{r}$$

with the number density of electrons $n_e(r)$ from the Preliminary Reference Earth Model. Electrons farther from the neutrino contribute less to the potential than the electrons that are closer.
Now, the new Hamiltonian from this interaction

$$H_{\mathrm{BSM}} = \mathrm{diag}(V_{\mathrm{BSM}}, -V_{\mathrm{BSM}}, 0)$$

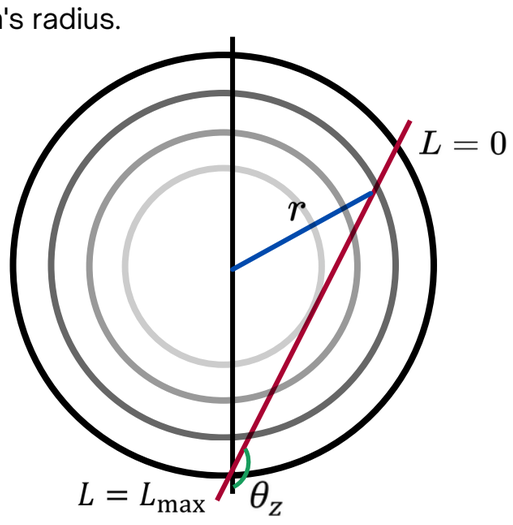
which is computed and solved numerically.

Set-up for detection

A detector is placed on the surface of the Earth. Neutrinos enter the Earth at zenith angle θ_z, propagating a distance

$$L_{max}(\theta_z) = a\sqrt{2(1 - \cos(2\theta_z - \pi))}$$

where a is the Earth's radius.



Along the neutrino's path, its distance from the Earth's center at a given point

$$r(L, \theta_z) = \sqrt{L^2 + a^2 - 2aL\cos(\pi - \theta_z)}$$

to evaluate the local electron density $n_e(r)$ using the PREM model for the interaction with matter and the new interaction.
The neutrino evolution is computed step-by-step along the trajectory by solving the Schrödinger equation numerically using the full Hamiltonian:

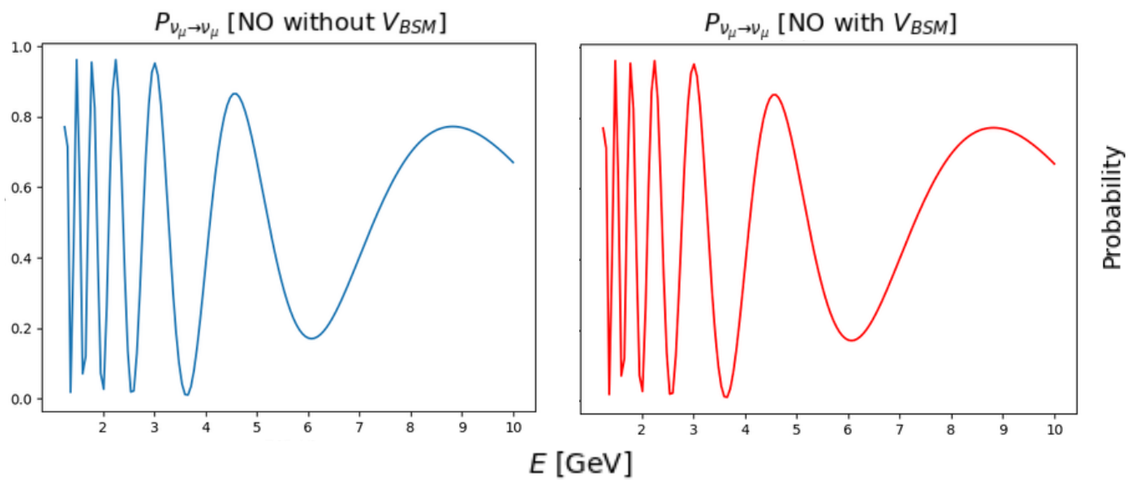
$$H_{\mathrm{total}} = H_{\mathrm{vac}} + H_{\mathrm{matter}} + H_{\mathrm{BSM}}$$

So, the computational implementation is:

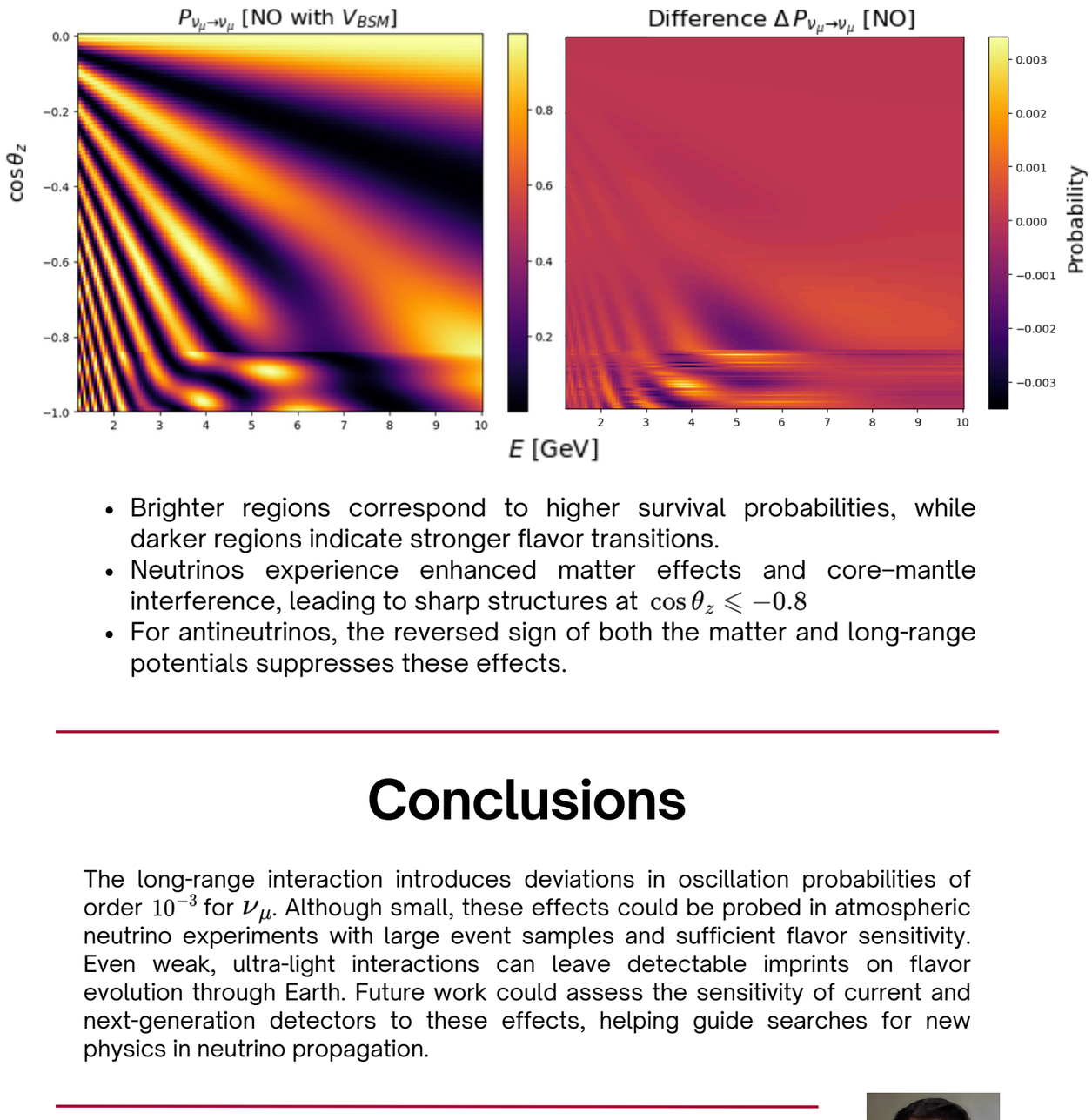
- The volume integral for the new interaction was computed using a **Simpson's algorithm** (arrays were defined for the integration to evaluate the 3D integral grid)

Oscillation Probabilites

Neutrinos change flavor as they travel, even in vacuum. When propagating through the Earth, interactions with electrons modify these transitions via the matter effect, enhancing or suppressing oscillations depending on energy and path. This effect becomes especially relevant for upward-going neutrinos crossing dense Earth layers.
To illustrate the energy dependence at fixed direction, survival probabilities for ν_μ are shown for a selected zenith angle:



The full oscillation pattern is shown next as a function of energy and zenith angle



- Brighter regions correspond to higher survival probabilities, while darker regions indicate stronger flavor transitions.
- Neutrinos experience enhanced matter effects and core–mantle interference, leading to sharp structures at $\cos\theta_z \leqslant -0.8$
- For antineutrinos, the reversed sign of both the matter and long-range potentials suppresses these effects.

Conclusions

The long-range interaction introduces deviations in oscillation probabilities of order 10^{-3} for ν_μ. Although small, these effects could be probed in atmospheric neutrino experiments with large event samples and sufficient flavor sensitivity. Even weak, ultra-light interactions can leave detectable imprints on flavor evolution through Earth. Future work could assess the sensitivity of current and next-generation detectors to these effects, helping guide searches for new physics in neutrino propagation.

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